

AlphaLoop

Autonomous Neural Options Trader

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The Problem & Our Solution

Problem

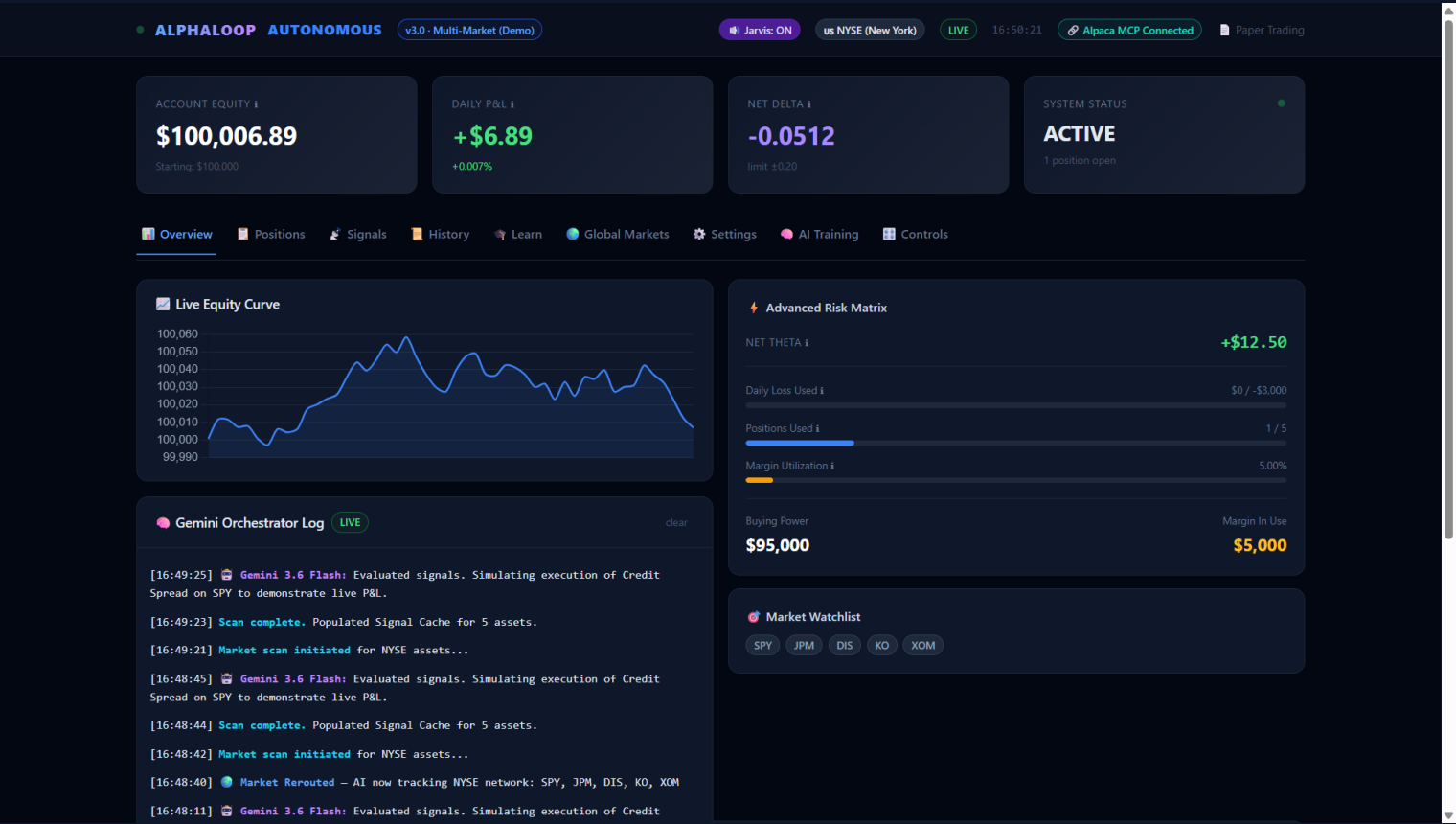
- Options trading demands reading RSI,
- IV Rank, Delta & Theta simultaneously.
- Humans miss signals. Algorithms lack judgment.

Solution

- A multi-agent swarm where each agent
- specialises in one layer of the pipeline.

Agents

- ■ Market Scanner — RSI · IV · Trend
- ■ Strategy Selector — Gemini 3.6 Flash
- ■■ Risk Guardian — Hard-coded Python rules
- ■ Execution Agent — Alpaca MCP



LLM Orchestration — Visible Reasoning

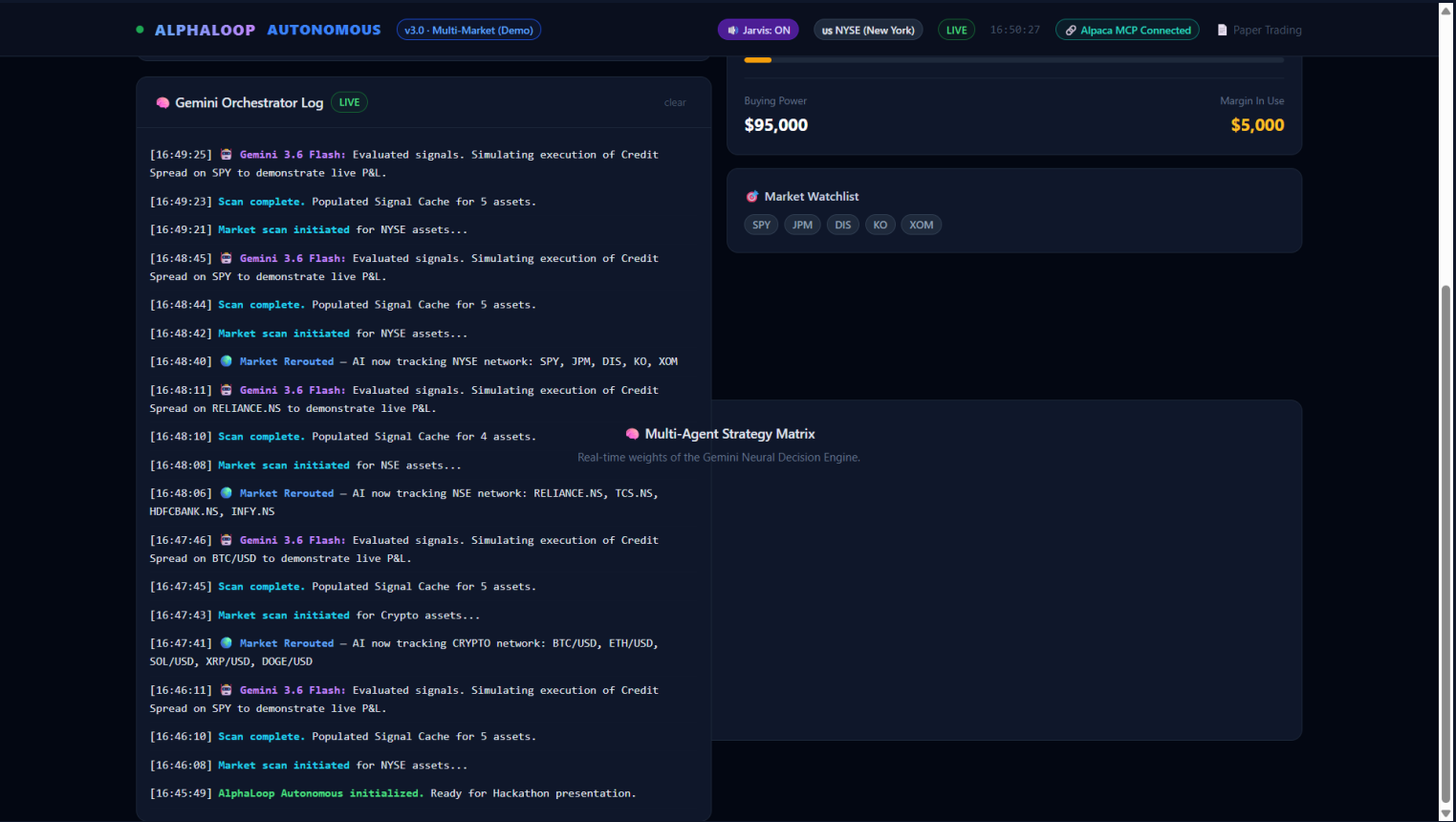
Gemini 3.6 Flash

- AI reads live RSI + IV Rank for each symbol.
- Formulates a plain-English financial thesis.
- Not a black box — reasoning appears in the UI.

■ Example reasoning output:

- "High IV Rank (72%) favors selling premium.
- Neutral RSI (49) → Iron Condor selected."

■ Selects: Iron Condor · Bull Put · Bear Call



Gemini model: gemini-3.6-flash | API: Google AI Studio

Risk Guardian — Deterministic Firewall

Zero-Trust Python Rules Engine

■ Hard limits that the LLM CANNOT override.

■ Max Portfolio Delta: 0.20

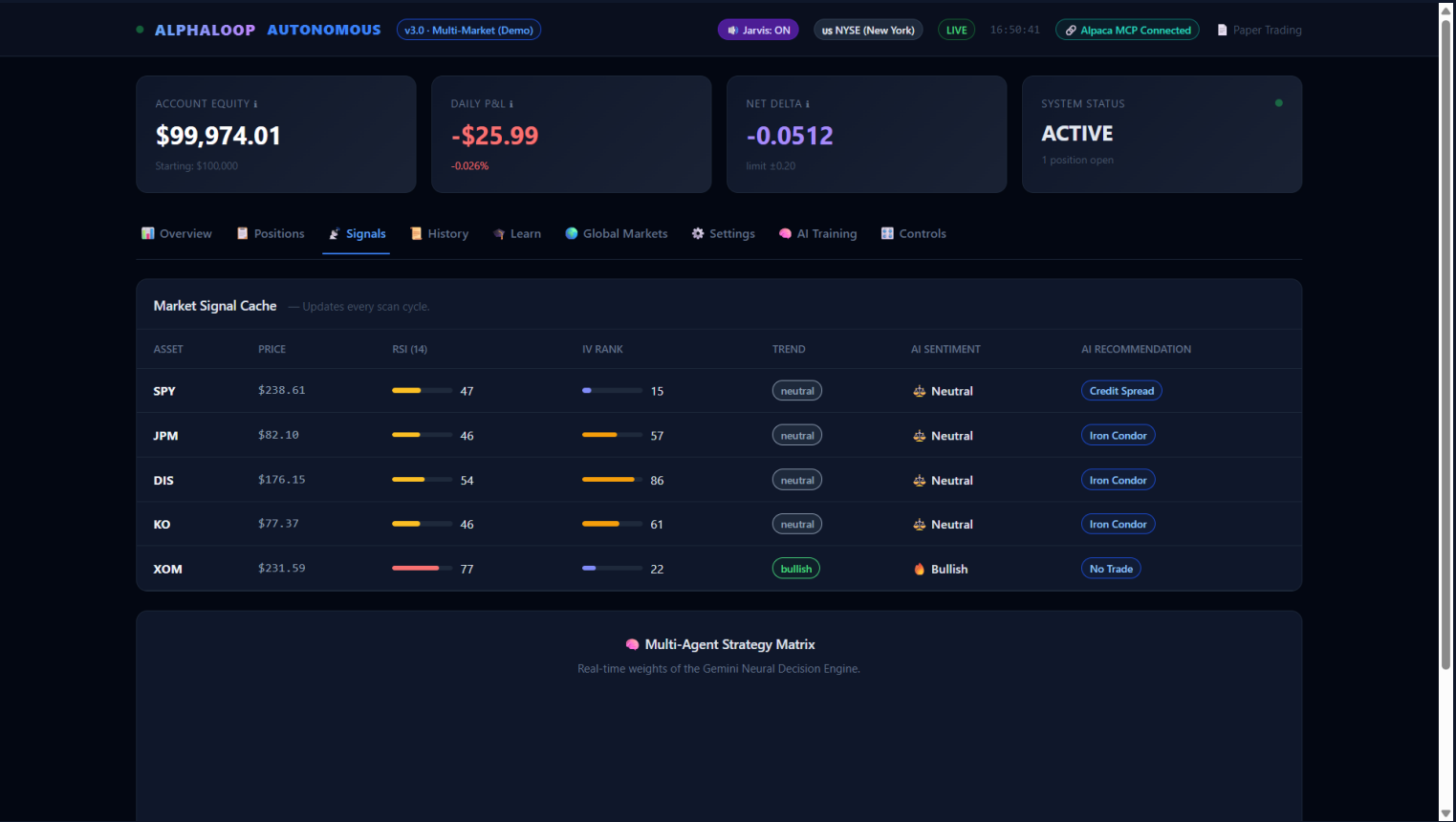
■ Max Single Position Size: 5% of buying power

■ Daily Loss Cap: 3%

■ If ANY rule is violated → trade REJECTED.

■ Automated test suite proves correctness.

■ (tests/test_risk.py)



AI proposes — code decides.

Model Context Protocol (MCP) Execution

Alpaca MCP Server Integration

■ Industry-first autonomous multi-leg MCP integration.

■ Connects to official alpaca-mcp-server.

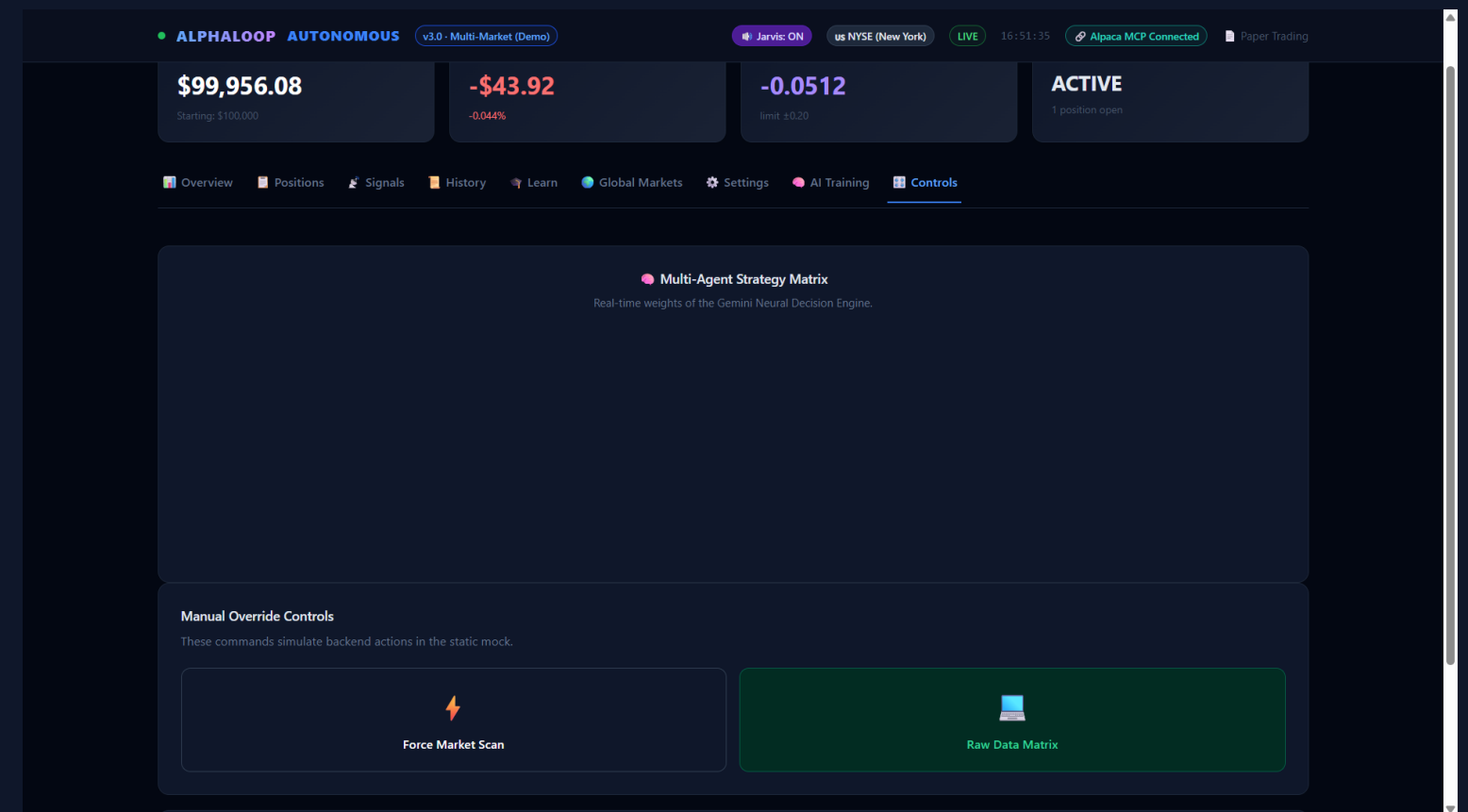
■ Calculates optimal strike prices dynamically.

■ Routes complex multi-leg orders to Alpaca:

■ Iron Condor (4 legs)

■ Bull Put Spread (2 legs)

■ Protective Put (2 legs)

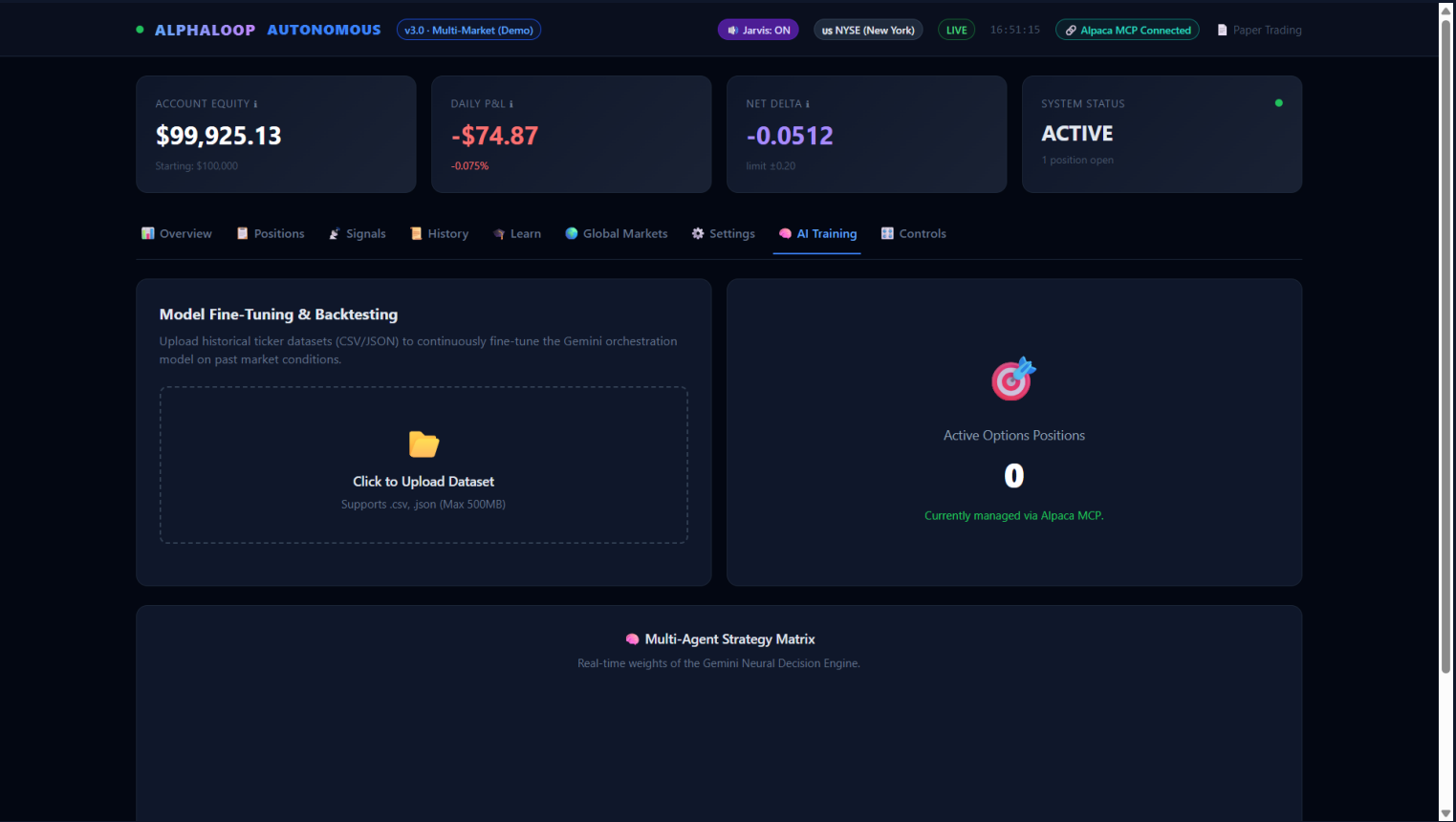


Execution proof visible live at alphaloop-autonomous.onrender.com/api/state

Deep Observability & Global Market Routing

Bloomberg-Style Terminal Dashboard

- Tracks 20 global market networks simultaneously.
- NYSE · NASDAQ · LSE · TSE · Crypto · ASX...
- Dark Pool block-trade tracker for institutional flow.
- Equity curve chart updated after every trade.
- Agent learning logs & internal state visible in UI.



Frontend: alphaloop-autonomous.vercel.app

Live Infrastructure & Links

All systems are LIVE right now

■ **Vercel Dashboard**
alphaloop-autonomous.vercel.app

■ **Render AI Backend**
alphaloop-autonomous.onrender.com/api/state

♥ **Health Check API**
alphaloop-autonomous.onrender.com/api/health

■ **YouTube Demo**
youtube.com/watch?v=EN-QTGJFPms

■ **GitHub Source**
github.com/shambhushekharsinha-engg/alphaloop-autonomous

Tech Stack

Python 3.11

FastAPI

Gemini 3.6 Flash

Alpaca MCP

Vercel

Render

AsyncIO

